- By 2026, generative design AI will automate 60% of the design effort for new websites and mobile apps - *Gartner*
- By 2027, nearly 15% of new applications will be automatically generated by AI without human intervention. This is not happening at all today. – *Gartner*
- According to Accenture's 2023 Technology Vision report, 97% of global executives agree that foundation models will enable connections across data types, revolutionising where and how AI is used.

The key characteristics of LLM technology are:

- LLMs can automate data cataloguing, enhancing speed and efficiency.
- They can continuously monitor and improve data quality and detect anomalies.
- These models can generate insights from both structured and unstructured data.
- LLMs enhance data quality control by detecting inconsistencies and anomalies. They can automate the process of checking data against predefined quality standards, making it faster and more efficient.
- They can complete a given text coherently, translate text between languages, and summarise concisely.
- LLMs perform various NLP tasks, understanding and processing human language, and allowing users to ask questions in a conversational manner. They can generate insights in natural language, making them accessible to non-technical stakeholders.

## LLM data preparation

Figure 1 shows the steps involved in LLM data preparation.

***Data requirements:*** Establish the AI strategy. Identify areas where LLMs add value and the types of data that are needed for training LLMs, and select the right tools and technology for GenAI and data management. Identify the various data source systems covering enterprise smarter applications, machine learning tools and real-time analytics. Categorise data into structured, semi-structured, and unstructured types. Structured data refers to databases of the enterprise. Unstructured data comprises videos, images, text messages, etc.

***Data collection:*** Data sources provide the insight required to solve business problems. The various data sources are web, social media, text documents, etc. Collected data is used for training LLMs.

For example, to perform sentiment analysis by the trained LLM, the collected data should include a large number of reviews, comments, and social media posts.

Web scraping is the automated method of extracting data from various websites. Crawlee and Apify Universal scrapers are examples of web scraping tools.

***Data organisation:*** Preprocess data by cleansing, normalising, and tokenising it.

- Data cleansing involves the identification and removal of inaccurate, incomplete, or irrelevant data. Duplicate data will be removed and incorrect data values will be fixed.
- Data normalisation transforms data to standard format for easy comparison and analysis.
- Data tokenisation breaks the text into individual words or phrases using natural language processing (NLP). Tokenisation happens at word level, character level or sub-word level.

***Data encoding:*** Feature engineering is crucial here. It involves creating features based on pre-processed data. Features are numerical representations of the text that the LLM can understand. The data is split, augmented, and encoded during this stage.

- Splitting is the process of dividing the data into training, validation, and testing sets. Training set is leveraged to train LLMs.
- Augmenting helps in synthesising new data and transforming the existing data.
- Encoding embeds data into tokens or vectors.

***Data storage:*** This step primarily involves the Model hub, blog storage, and databases. The Model hub consists of trained and approved models that can be provisioned on demand, acting as a repository for model checkpoints, weights, and parameters. Comprehensive data architecture covering both structured and unstructured data sources is defined

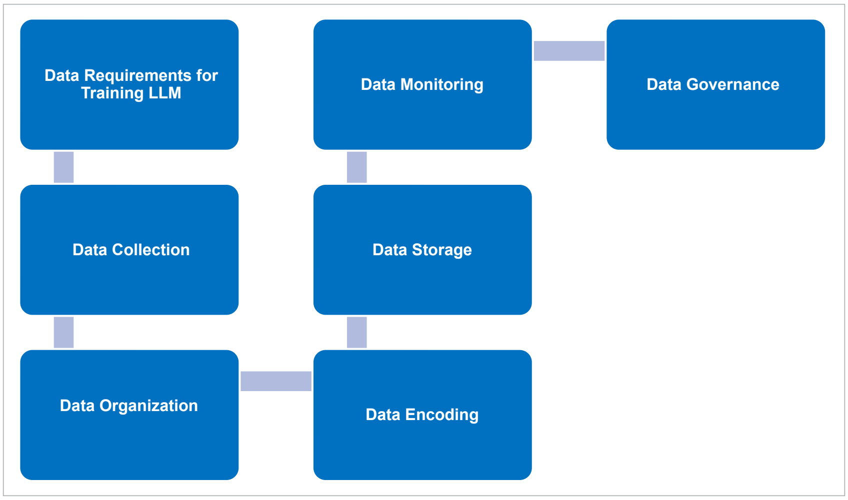


Figure 1: Data preparation for LLM